

RAM, GPU and Storage Troubleshooting

Assignment Answers

1. RAM reseating: procedure and observation record

Safety: Shut down the computer, unplug the power cable and peripherals, and hold the power button briefly to discharge residual power. Use anti-static precautions. Never remove RAM while powered. For laptops, follow the service manual; some models have soldered RAM or internal batteries.

Step 1: Shut down the PC and disconnect power. For a laptop, unplug the charger and follow the maker's battery-disconnection instructions.

Step 2: Open the case or approved memory panel with the correct tool. Keep screws organized.

Step 3: Locate the RAM modules and note their slot positions and retaining clips.

Step 4: Release the clips as designed and lift the module by its edges. Do not force it or touch the gold contacts.

Step 5: Check for visible damage, dust, or broken clips. Avoid abrasive cleaning methods.

Step 6: Align the module notch with the slot key and press evenly until it is fully seated and the clips lock.

Step 7: Close the case, reconnect power, and start the PC. Check that it boots and reports the expected memory amount.

Example observation (template only): The RAM was reseated, no visible damage was seen, clips locked, and the PC started normally. Replace this with your actual observations. I have not physically opened or tested your computer.

2. Gaming PC GPU troubleshooting checklist

1. Record the symptom: no display, artifacts/flickering, game crashes, low frame rate, or black screen under load.
2. Check monitor power/input and display cable at both ends. Try a known-good cable or monitor if available.
3. With a discrete GPU, ensure the monitor cable is connected to the graphics card's outputs, not the motherboard video port.
4. Power off and unplug the PC. Check that the GPU is seated in its PCIe slot and required GPU power connectors are firmly attached. Never open the PSU.
5. Check GPU fans, but remember some cards stop their fans at low temperatures; fan inactivity alone does not prove failure.
6. On Windows, check Device Manager for the display adapter, warning icons, or error codes. If detected, install/reinstall the correct driver from the GPU maker.
7. Check temperatures and behavior during a normal game/test. Stop if there is a burning smell, severe overheating, or repeated power loss.
8. If possible, test with a known-good compatible GPU or test the suspect GPU in another compatible PC.
9. Repeat the original workload and confirm the display issue, crashes, or detection errors no longer occur.

3. Simulating a SATA or NVMe storage connection issue

Use only a non-critical test system with backups. Never disconnect/reconnect an internal drive while powered unless the hardware is specifically designed for hot-swap. Shut down and unplug first. For SATA, reseat both data and power cables. For M.2 NVMe, follow the motherboard manual to remove the heatsink if needed, reseat the module at the correct angle, and secure it without overtightening.

After reconnecting, enter BIOS/UEFI and check whether the drive model/capacity appears, then try booting the operating system.

Stage	Possible message/indicator	Meaning
Disconnected	Drive absent from BIOS; “No bootable device” or “Operating system not found.”	Firmware cannot detect the drive or find boot files; wording varies by system.
Reconnected successfully	Drive model/capacity appears in BIOS and the OS boots.	A loose connection may have been the cause. Confirm stability and back up files.
Still failing	Drive absent or a drive/SMART error appears.	Possible bad cable/port, failed drive, incorrect seating, or configuration issue.

Report note: These are expected examples, not results observed on a real test system. Record the exact message and drive details from your own practical.

4. PC randomly restarts: component to check first

Starting point: RAM, if there are no clear overheating or power-related signs. Faulty or poorly seated memory can cause instability and restarts, and is straightforward to reseat and test. This is only a first check—not a diagnosis; overheating, PSU, GPU, motherboard, storage, or OS issues can also cause restarts.

- Note when restarts occur: during gaming, at idle, during boot, or after moving the PC. If there is a burning smell or severe heat, shut down and stop.
- Power off, unplug, use ESD precautions, and reseat the RAM.
- Test one module at a time in the motherboard-recommended slot; test modules separately to identify a bad stick or slot.
- Run the built-in memory diagnostic or a reputable bootable memory test. Repeatable errors suggest further testing or compatible RAM replacement.
- If memory passes, check CPU/GPU temperatures and cooling, GPU seating/power, system logs, and PSU output with suitable equipment or a technician. Back up data if storage errors appear.

Conclusion: Record actual observations rather than claiming a test was performed if it was not. Disconnect power before reseating components, protect against static discharge, and seek professional help if unsure.